

SA03 DEVELOP LEGIONELLA MANAGEMENT PLAN | O (MAX: 1 PT)

Intent: Implement protocols to reduce risk of *Legionella* colonization.

Summary: This feature requires projects to manage recirculating hot water systems against *Legionella* colonization.

Issue:

All water systems require some degree of validation to ensure that health targets are met under the operational conditions, for which sampling is an effective verification tool.²²⁷ *Legionella* bacteria is naturally present in waters at low concentrations, but it may colonize recirculated water systems and can cause lung disease and even death if contaminated water aerosols are inhaled.²²⁸ Legionnaire's disease affects especially the immunocompromised, smokers and those over age 50.²²⁸ Assets commonly vulnerable to *Legionella* infestation include domestic potable and hot water systems, cooling towers, humidifiers, misters, decorative fountains, spas and hot tubs.²²⁸

Solutions: Minimizing the risk of *Legionella* requires a thorough and quantitative risk assessment of buildings' water assets, identifying locations where control is required, and a well-documented maintenance and operations program.²¹³ Implementing a proper *Legionella* management plan should reduce the risk of exposure to pathogenic bacteria.

Part 1 Part 1 (Max: 1 Pt)

For All Spaces:

The project provides a *Legionella* management plan that meets the following requirements:

- a. Addresses hot water systems, cooling towers, decorative fountains and any other devices or spaces under control of the project where water is recirculated and aerosolized.
- b. Includes the items listed below:
 1. Determination of roles for *Legionella* management in the building, distinguishing those under project control from those that may be the responsibility of building management or other parties.
 2. Water system inventory and process flow diagrams of systems within the project boundary.
 3. Hazard analysis of water assets within the project boundary. If the project does not operate the building hot water supply system (e.g., boilers, heaters, pumps or hot water risers), then an explanation of the building-wide *Legionella* management policies (if any) and how they influence risk is included.
 4. A list of monitoring actions for relevant variables (e.g., temperature or residual chlorine), performance limits associated with these variables and corrective actions when variables exceed such limits.
 5. A list of critical control points (locations where actions to maintain relevant variables listed in (4) within performance limits are applied) within the project boundary.
 6. Verification and validation procedures for evaluating the suitability and proper implementation of the management plan. A *Legionella* sampling schedule is included if projects have operational control over cooling towers and spas.
 7. Protocols for documenting results of monitoring activities and corrective actions. If sampling for *Legionella* is planned, results are included.

The following requirement is met:

- a. Documentation of monitoring results, corrective actions and *Legionella* sample results as stated in the *Legionella* management plan are submitted annually through the WELL digital platform.